

Test 1 · Algebra

Grade 8

40 minutes 40 marks 21 questions No calculator

- 1–3 Revision of the Grade 7 course
- 4–5 Algebraic expressions
- 6–8 Algebraic fraction. Cancelling fractions

Algebra 8, p. 3 · Stage 8 review

Algebra 8, §1 · Stage 9 · 2.1–2.2

Algebra 8, §2 · Stage 9 · 2.5

- Answer every question. Show your working — method carries marks.
- Whenever you cancel a fraction, state the permissible values of the variable.
- No calculator.

What the paper weighs

TOPIC	MARKS	SHARE
Lessons 1–3 · Revision of the Grade 7 course	14	35%
Lessons 4–5 · Algebraic expressions	9	22%
Lessons 6–8 · Algebraic fraction. Cancelling fractions	17	42%

Cancelling carries the largest share because every question in Section C also exercises the factorising from lessons 1–3.

A Section A · Short answers

EASY

10 marks

One mark each. A single line of working is enough.

1 Simplify: $x^6 \cdot x^4$ [1]

ANSWER

$$x^{10}$$

WORKING

Same base, so the indices add: $6 + 4 = 10$.

2 Simplify: $(b^4)^3$ [1]

ANSWER

$$b^{12}$$

WORKING

A power of a power multiplies the indices: $4 \times 3 = 12$.

Watch for b^7 — adding where it should multiply.

3 Expand: $4(3x - 2)$ [1]

ANSWER

$$12x - 8$$

WORKING

Multiply both terms inside the bracket by 4.

4 Factorise: $10a - 15$ [1]

ANSWER

$$5(2a - 3)$$

WORKING

The common factor is 5.

5 Which two of these are fractional expressions?

$$\frac{2x}{7} \quad \frac{4}{x} \quad \frac{x}{3} - 5 \quad \frac{9}{x+2} \quad [1]$$

ANSWER

$$\frac{4}{x} \text{ and } \frac{9}{x+2}$$

WORKING

An expression is fractional when a *variable* stands in a denominator. In $\frac{2x}{7}$ and $\frac{x}{3} - 5$ the denominator is a number, so those are whole expressions.

6 State the permissible values of $\frac{1}{x-6}$ [1]

ANSWER

$$x \neq 6$$

WORKING

The denominator must not be zero: $x - 6 \neq 0$.

7 Find the value of $\frac{x+5}{x-2}$ when $x = 3$ [1]

ANSWER

$$8$$

WORKING

$$\frac{3+5}{3-2} = \frac{8}{1} = 8.$$

8 Cancel: $\frac{8a}{12a}$ [1]

ANSWER

$$\frac{2}{3}, a \neq 0$$

WORKING

Divide numerator and denominator by $4a$.

9 Cancel: $\frac{15x^3}{5x}$ [1]

ANSWER

$3x^2, x \neq 0$

WORKING

Divide numerator and denominator by $5x$.

10 Cancel: $\frac{5(x-2)}{10(x-2)}$ [1]

ANSWER

$\frac{1}{2}, x \neq 2$

WORKING

The bracket $(x-2)$ is a factor of both, so it cancels, and $\frac{5}{10} = \frac{1}{2}$.

B Section B • Two marks each

Method marks are available even when the final answer is wrong.

MEDIUM

14 marks

11 Simplify: $(3a^3b^2)^2$ [2]

ANSWER

$9a^6b^4$

WORKING

Square every factor: $3^2 = 9$, $(a^3)^2 = a^6$, $(b^2)^2 = b^4$.

A common slip is $3a^6b^4$ — the 3 must be squared as well.

12 Expand and simplify: $(x+5)(x-2) - x^2$ [2]

ANSWER

$3x - 10$

WORKING

$(x+5)(x-2) = x^2 + 3x - 10$; subtracting x^2 leaves $3x - 10$.

13 Factorise: $9x^2 - 16y^2$ [2]

ANSWER

$(3x-4y)(3x+4y)$

WORKING

A difference of two squares, since $9x^2 = (3x)^2$ and $16y^2 = (4y)^2$.

14 Cancel: $\frac{3x + 12}{x^2 - 16}$ [2]

ANSWER

$$\frac{3}{x - 4}, x \neq 4, x \neq -4$$

WORKING

Factorise first: $\frac{3(x + 4)}{(x - 4)(x + 4)}$. The factor $(x + 4)$ cancels.

15 State the permissible values of $\frac{x + 2}{x^2 - 25}$ [2]

ANSWER

$$x \neq 5, x \neq -5$$

WORKING

$x^2 - 25 = (x - 5)(x + 5)$, which is zero at $x = 5$ and at $x = -5$.

16 Find the value of $\frac{a^2 - b^2}{a + b}$ when $a = 9, b = 4$ [2]

ANSWER

$$5$$

WORKING

Cancel before substituting: $\frac{(a - b)(a + b)}{a + b} = a - b = 9 - 4 = 5$.

Substituting first gives $\frac{65}{13}$, which is also 5 — but cancelling first is the skill being tested.

17 For which value of x is $\frac{x - 7}{x + 1}$ equal to zero? [2]

ANSWER

$$x = 7$$

WORKING

A fraction is zero when its numerator is zero and its denominator is not: $x - 7 = 0$ gives $x = 7$, and $x = 7$ is permissible because $x \neq -1$.

An answer of $x = 7$ or $x = -1$ shows the two conditions have been confused.

C Section C • Four marks each

HARD

16 marks

Factorise before you cancel, and state the values the variable may not take.

- 18 (a) Factorise completely: $x^3 - 9x$ [2]

(b) Hence cancel $\frac{x^3 - 9x}{x^2 + 3x}$ and state the permissible values. [2] [4]

ANSWER

(a) $x(x - 3)(x + 3)$ (b) $x - 3$, $x \neq 0$, $x \neq -3$

WORKING

(a) Take out x first, giving $x(x^2 - 9)$, then the difference of two squares gives $x(x - 3)(x + 3)$.

(b) The denominator factorises as $x(x + 3)$, so $\frac{x(x - 3)(x + 3)}{x(x + 3)} = x - 3$. That denominator is zero at $x = 0$ and at $x = -3$, so both values are excluded.

“Completely” means going on until nothing factorises further; stopping at $x(x^2 - 9)$ loses a mark.

- 19 Cancel $\frac{x^2 - 8x + 16}{x^2 - 16}$, state the permissible values, then find the value of your answer when $x = 9$.

[4]

ANSWER

$\frac{x - 4}{x + 4}$, $x \neq 4$, $x \neq -4$; at $x = 9$ it is $\frac{5}{13}$

WORKING

The numerator is a perfect square: $x^2 - 8x + 16 = (x - 4)^2$. The denominator is $(x - 4)(x + 4)$. One factor of $(x - 4)$ cancels and leaves $\frac{x - 4}{x + 4}$.

At $x = 9$: $\frac{9 - 4}{9 + 4} = \frac{5}{13}$.

- 20 (a) Cancel: $\frac{2x^2 - 2y^2}{3y - 3x}$ [3]

(b) Explain in one sentence why $\frac{x + 5}{5}$ cannot be cancelled. [1] [4]

ANSWER

(a) $-\frac{2(x + y)}{3}$, $x \neq y$ (b) because the 5 is a term, not a factor

WORKING

(a) $2x^2 - 2y^2 = 2(x - y)(x + y)$ and $3y - 3x = -3(x - y)$. The swap $y - x = -(x - y)$ is what makes the two factors match; cancelling $(x - y)$ leaves $-\frac{2(x + y)}{3}$.

(b) In $x + 5$ the 5 is a *term*, not a *factor*, and only a factor common to the whole numerator and the whole denominator may be cancelled.

Part (b) separates students who follow the rule from students who know why it holds.

- 21 Cancel $\frac{2x^2 + 5x - 3}{x^2 - 9}$, stating the permissible values. [4]

ANSWER

$$\frac{2x - 1}{x - 3}, \quad x \neq 3, x \neq -3$$

WORKING

The numerator factorises as $(2x - 1)(x + 3)$ — check by expanding: $2x^2 + 6x - x - 3 = 2x^2 + 5x - 3$. The denominator is $(x - 3)(x + 3)$, so the factor $(x + 3)$ cancels.

END OF MARK SCHEME